

SplitAlign: Token-Level Cross-Lingual Difference via Aligned LLM Judgments

Hack Apertus Online 2026 · Track 2A UZH SwissGov-RSD · Waterloo Agent Lab · Sharon Basovich · github.com/sharonbasovich/splitalign · Apache-2.0 code / CDLA-Permissive-2.0 predictions / CC-BY-4.0 report

1 Method

SplitAlign decomposes token-level cross-lingual semantic-difference scoring (de/fr/it Swiss administrative document pairs) into auditable stages, preserving the original whitespace-token indices end-to-end so outputs land exactly on the official token label space:

- **Segment** — sentence/paragraph segmentation that records each segment's token offset.
- **Align** — monotone dynamic-programming alignment of cross-lingual sentences with 1:0, 0:1, 1:1, 1:2 and 2:1 (merge) operations, scored by batched Apertus similarity under a **banded candidate policy** ($|i/(n-1) - j/(m-1)| \leq 0.2$ or ± 2 absolute): far-off-diagonal pairs score 0 without an API call — measured 304,968 → 108,260 candidate pairs (35%) on the dev corpus.
- **Judge** — short structured-JSON Apertus judgments on aligned pairs: graded semantic difference 0–5 plus verbatim differing spans, with repair/salvage for malformed responses and explicit handling of repeated tokens, offsets, punctuation, boilerplate and end-of-document omissions.
- **Score** — sentence scores plus span boosts projected onto the exact token arrays; asymmetric additions/omissions get explicit labels.

A whole-document token-annotation prompt serves as the **re-implemented baseline** (same response contract; not the upstream template).

2 Apertus use (required component)

CSCS hosted inference `api.inference.cscs.ch/v1`, model `swiss-ai/Apertus-v1.5-8B` (non-thinking), temperature 0, fixed seed. Apertus performs both the similarity scoring and the graded judgments — no fine-tuning or hidden-state access is assumed. Every call is cached on backend|model|prompt-version|kind|split|item|payload keys and logged redacted (hashes, token counts, latency only). A shared budget caps NEW requests *and* tokens across both methods at the **outbound-attempt boundary**, so retries and failures cannot bypass the cap; identical cache hits are free.

3 Data, licensing, and the held-out firewall

Only `dev/train` and `dev/val` of `ZurichNLP/swissgov-rsd` (CC-BY-4.0 per its Hugging Face card) are fetched — at a pinned commit through an allowlist + fail-closed guard (disallowed splits/paths/symlinks and unmanifested IDs abort). The test split is never fetched, read, or evaluated in this build. Metrics follow the official token-level Spearman convention including its `-1` punctuation filter and `list_to_drop.txt` exclusion (reported both ways). All scoring code is **original Apache-2.0** written from the metric specification — no vendored upstream code (upstream ships no repo-level license).

4 Evaluation — dev/val, 10 documents/language (exploratory; baseline incomplete)

Run `20261001T211451Z-8de1aab7`: 30/30 planned SplitAlign items, 97 new requests / 72.6k new tokens / 159 s on top of cache hits from an earlier partial run (24/30 docs, 499 requests / 600k tokens, token cap hit). Strict macro Spearman = mean over de/fr/it, null if any language is undefined; nonfinite values serialize as JSON null with explicit reasons. **The re-implemented whole-document Apertus baseline was declared in the run manifest but produced no output in this scope, so no matched same-ID Apertus baseline exists** (the viewer marks the run PARTIAL). Earlier pre-fix diagnostics are quarantined in `results/runs/legacy-pre-runscope/`; mock outputs are plumbing checks only.

Offline same-ID context (`scripts/encoder_context_dev.py`, no API calls): the dataset authors' published DEV encoder predictions (upstream commit `1807a42`, `data/evaluation/encoder_predictions/dev/`; every ID validated against our dev manifest; files not redistributed — URLs/sha256 in `results/encoder_context/`) scored on exactly the same 30 IDs with the same metric code. Comparator fixed in advance: EuroBERT-210m reference, all four mmBERT parallel DiffAlign variants as context. Paired = document-level paired bootstrap of the strict-macro difference (SplitAlign – system), B=2000.

System (same 30 dev/val IDs)	de	fr	it	Strict macro	Paired diff [95% CI]
SplitAlign (Apertus-8B, this run)	0.254	0.267	0.224	0.248	—
EuroBERT-210m DiffAlign (reference)	0.055	0.196	0.198	0.150	+0.099 [–0.008, +0.184]
mmBERT parallel_en-de	0.180	0.230	0.266	0.225	+0.023 [–0.073, +0.102]
mmBERT parallel_en-sw	0.170	0.228	0.271	0.223	+0.025 [–0.070, +0.104]
mmBERT parallel_en-de-fr-it	0.199	0.243	0.316	0.253	–0.004 [–0.103, +0.077]
mmBERT parallel_all_1epoch	0.206	0.243	0.299	0.249	–0.001 [–0.095, +0.084]

Honesty note. Exploratory dev-only figures on a 30-document subset: numerically above the preregistered EuroBERT reference but the paired CI includes zero, and indistinguishable from the stronger mmBERT variants — **no improvement over published encoders is claimed**, no held-out claim is made, and subset figures are not comparable to whole-dev (168 docs/lang) numbers. Paper prior-work figures are not reproduced here.

5 Reproducibility

Repo github.com/sharonbasovich/splitalign (public); exact code commit is given on the submission form. Root make run builds Docker and runs a mock pipeline on a clean checkout; real runs set `SPLITALIGN_BACKEND=apertus + APERTUS_*` env vars (contradictory config fails closed). Every run writes an immutable `results/runs/<ts-id>/` scope — predictions, intended-ID manifest, details, redacted call log, per-mode and matched-ID evals — committed to git. Seed 0 throughout; payload-keyed cache gives bit-identical reruns. New run manifests record source commit/dirty state, prompt and package version, gold sha256, score config and caps (the 30-doc run predates this field: code state 7ad62c3..8480423, left null, not back-filled). The static bilingual viewer (`track_2a/viewer/`) is exported from an explicit `--run` only and shows run id, intended vs produced modes, coverage and cap/partial status with real-vs-mock banners.

7 AI-assistance disclosure & references

Strategy brief and implementation produced by AI assistants (reviewer session + Devin); S. Basovich supplied goals, approvals and credential provisioning. Sources: Hack Apertus Track 2A brief; ZurichNLP/SwissGov-RSD; HF dataset card (CC-BY-4.0); CSCS Inference API docs; [swiss-ai/Apertus-v1.5-8B](https://github.com/swiss-ai/Apertus-v1.5-8B); published dev encoder predictions from ZurichNLP/SwissGov-RSD@1807a42 (context only).

6 Limitations

Uniform token scores per aligned sentence except reported spans; monotone alignment only (no reordering); banded candidates never score far-off-diagonal pairs; judgments may omit end-of-document differences (handled, logged); held-out integrity is a development guard, not a proof.